

CSA5112 - Cryptography and Network Security

10. Triple DES (3DES)

Aim: To implement the Triple Data Encryption Standard (3DES) algorithm for encrypting and decrypting a plaintext message using a secure secret key.

Algorithm:

1. Start.
2. Import the required libraries.
3. Read the plaintext from the user.
4. Create a valid 24-byte Triple DES key.
5. Create a Triple DES cipher object in ECB mode.
6. Encrypt the plaintext.
7. Display the encrypted ciphertext.
8. Decrypt the ciphertext.
9. Display the original plaintext.
10. Stop.

Code:

```
from Crypto.Cipher import DES3

from Crypto.Util.Padding import pad, unpad

import base64

# 24-byte Triple DES key

key = b'123456789012345678901234'

# Create cipher

cipher = DES3.new(key, DES3.MODE_ECB)

# Input

plaintext = input("Enter Plain Text: ")

# Encrypt

encrypted = cipher.encrypt(pad(plaintext.encode(), DES3.block_size))
```

```
ciphertext = base64.b64encode(encrypted).decode()

print("\nEncrypted Text:", ciphertext)

# Decrypt

decrypted = unpad(cipher.decrypt(encrypted), DES3.block_size)

print("Decrypted Text:", decrypted.decode())
```

Sample Input:

```
Enter Plain Text: HELLO
```

Sample Output:

```
Enter Plain Text: HELLO WORLD

Encrypted Text: zS1sYk0LkzH0J005N6z8Hg==
Decrypted Text: HELLO WORLD
```

Result:

The Triple DES (3DES) algorithm was successfully implemented. The plaintext was encrypted using the Triple DES algorithm and successfully decrypted back to the original plaintext.